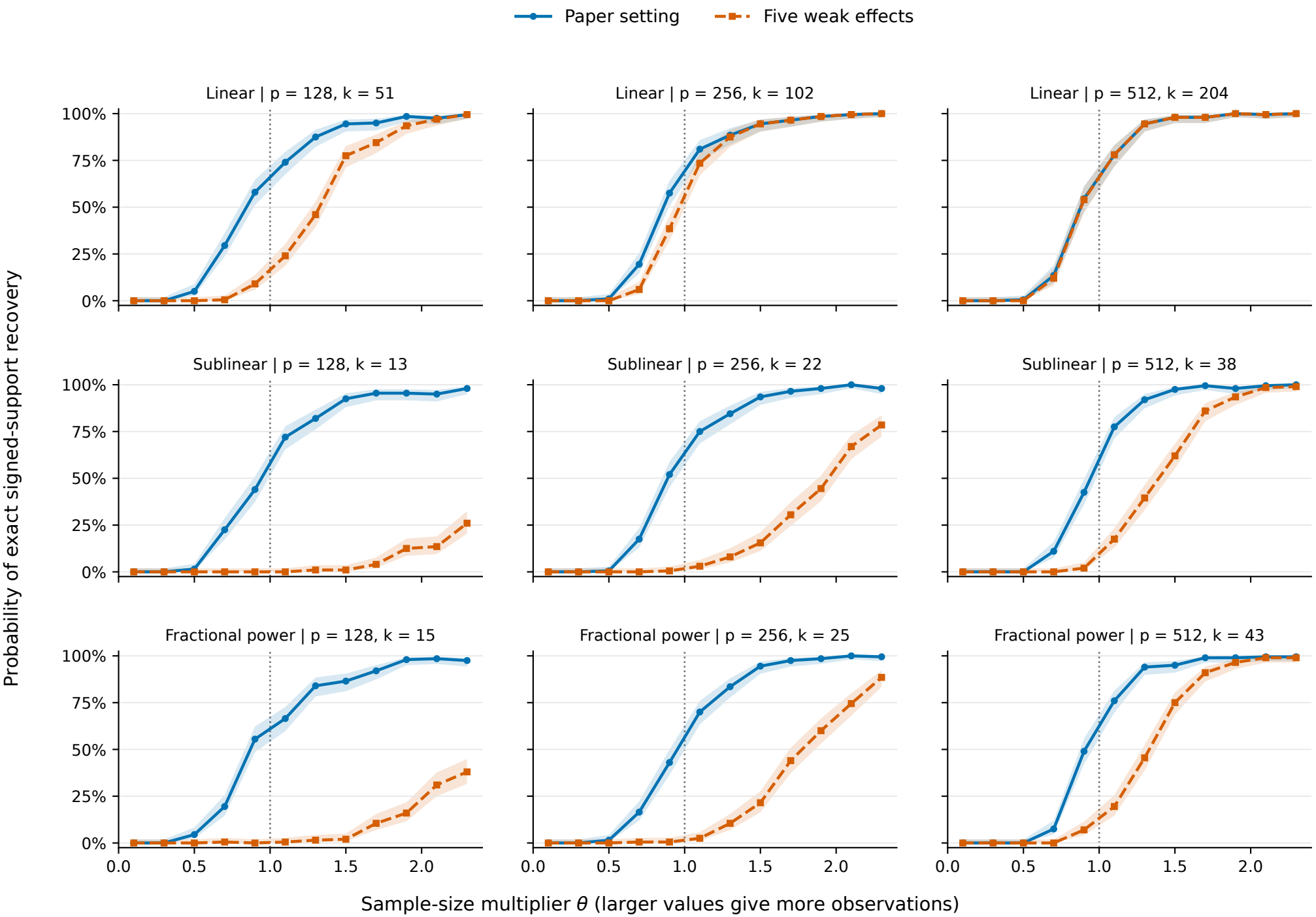
Exact Lasso recovery with strong and weak effects

Independent Gaussian predictors | Noise SD = 0.5



Paper setting: all active magnitudes = 1. Weak setting: five active magnitudes = 0.25; the rest = 1.
Points: 200 trials per condition. Shading: 95% pointwise Wilson Monte Carlo intervals.
Recovery requires every support location AND sign to be correct (estimated $|\beta| \le 10^{-7}$ counts as zero).
n = floor[2θk log(p − k)] + k + 1; random support and ± signs; natural logs. Sparsity uses α = 0.4, γ = 0.75.
Dotted line: asymptotic identity threshold, θ = 1. These are large-sample references, not finite-sample guarantees.